Understanding Neural Networks from the Ground Up: A NumPy-Only Implementation Compared with a Framework Baseline - Update 1

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***Abstract*—Neural networks are widely used to build machine-learning models. A typical complete neural network has many inputs, neurons, outputs, weights, and biases. For beginners, a simpler neural network with incomplete features can be a useful starting point for learning how to build a complete network [1].**

I. Introduction

We build a neural network with one neuron, few inputs, and a limited error-correction and learning mechanism. This is an initial step toward developing a complete neural network. Using the Wisconsin Breast Cancer dataset [2], we limit the amount of data used and keep the weights and other mechanisms simple. We implement a forward pass and train the model over multiple epochs to observe its behavior.

II. Theory

Fig. 1 shows the single neuron used in this work. It first computes the weighted sum of its inputs,

$$z = \sum_{i=1}^{n} w_i x_i, \tag{1}$$

and then applies the sigmoid activation to produce the prediction,

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}. \tag{2}$$

The prediction is compared with the real value y using the squared error,

$$L = \frac{1}{2}(\hat{y} - y)^2. \tag{3}$$

The difference between the prediction and the real value is

$$\delta = \hat{y} - y, \tag{4}$$

and the weight delta, which measures the effect of each weight on the loss, is

$$\Delta w_i = \delta \, x_i. \tag{5}$$

Each weight is then adjusted by gradient descent with learning rate α, as illustrated in Fig. 2:

$$w_i \leftarrow w_i - \alpha \, \Delta w_i. \tag{6}$$

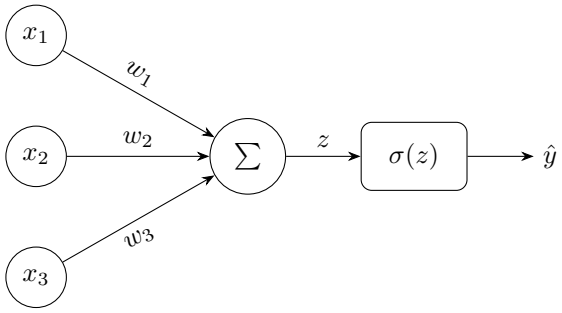


Fig. 1. A single neuron with three inputs. The weighted sum z is passed through a sigmoid activation to produce the output $\hat{y}$.

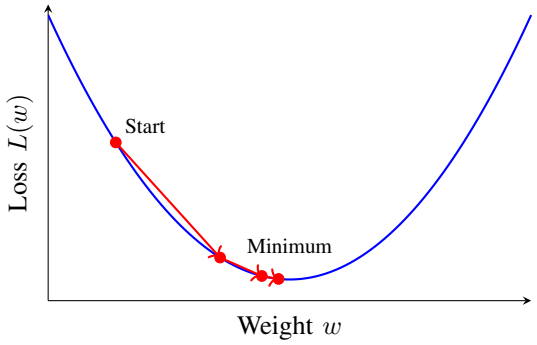


Fig. 2. Gradient descent on a loss curve. Each step moves w against the gradient, $w \leftarrow w - \alpha \, \Delta w$, toward the minimum.

III. Implemented Features

1) A forward-pass mechanism calculates the weighted sum of inputs and their associated weights, applies a nonlinear activation function, and produces the neuron's activation [1].
2) A neural-network function generates predictions from input and weight parameters using an activation function [1].
3) Training is performed over multiple epochs, processing each set of inputs in every epoch.
4) The derivative dL/dw is used to measure the effect of weights on the loss, and the weights are adjusted using the corresponding update formula; gradient-based learning and backpropagation are foundational neural-network methods

[3].

IV. Limitations

1) A single neuron behaves similarly to a linear model and does not provide deep learning.
2) With only one activation layer, the model may struggle to represent more complex nonlinear relationships [1].
3) Hard-coded steps may cause weights to overshoot or undershoot suitable values.

V. Upcoming Improvements

Future work will increase the number of inputs and neurons and introduce multiple hidden layers. Biases will be added to input-to-neuron and neuron-to-neuron connections. Backpropagation and optimization methods will be implemented to improve loss calculation and weight and bias updates [1], [3], supporting more accurate predictions as the model develops.

References

[1] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

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[3] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, pp. 533–536, 1986.